Do Now · retrieval

Lesson 137 · Third Normal Form: Removing Transitive Dependencies

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a partial functional dependency?

Q2

What does ROM stand for?

Q3

What are the outputs of a half adder when A=1 and B=1?

Q4

What is a composite key?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Check 2NF first — Remove repeating groups and partial dependencies.

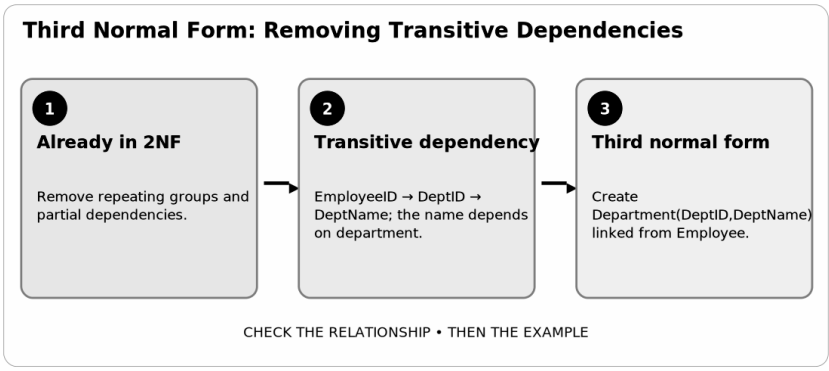
Core idea 2

Detect indirect dependency — EmployeeID → DeptID → DeptName; the name depends on department.

Core idea 3

Decompose tables — Create Department(DeptID,DeptName) linked from Employee.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

3NF — No transitive dependency of non-prime attributes on candidate keys.

transitive dependency — An attribute depends on a key indirectly through another non-key attribute.

functional dependency — Value of one attribute determines another.

update anomaly — Duplicate facts can disagree after an edit.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Employee(EmpID,EmpName,DeptID,DeptName) is in 2NF. Why may it violate 3NF?

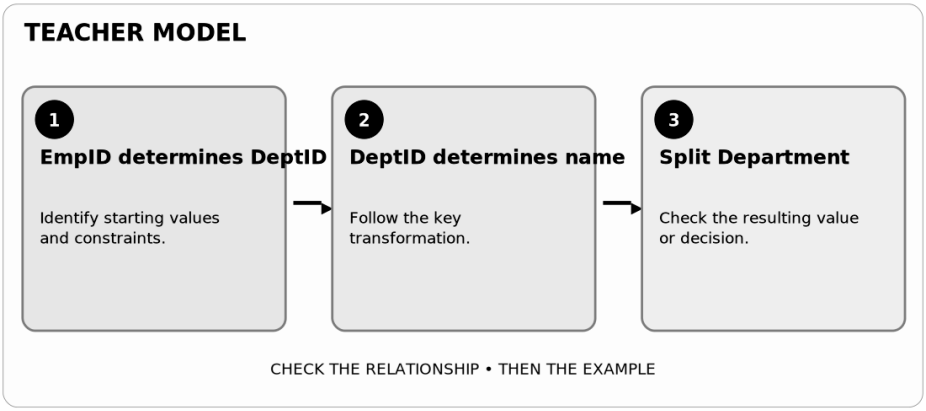
Follow the worked method

1. EmpID determines DeptID — Find the given data and name the required result.

2. DeptID determines name — Explain why this intermediate stage is necessary.

3. Split Department — EmpID identifies employee. DeptName depends on DeptID, not directly on EmpID, producing a transitive dependency. Split Employee(EmpID PK,EmpName,DeptID FK) and Department(DeptID PK,DeptName).

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Order identifies customer
Find the given data and name the required result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define transitive dependency and give a practical example. [6]

Question 2

Normalise Device(DeviceID,RoomID,RoomName) to 3NF, assuming RoomID determines RoomName. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why normalisation can reduce anomalies but may require additional joins. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In 3NF, a non-key value should be copied into every child row merely for convenience.

Use key vocabulary from memory

1. Explain 3NF in your own words.

2. Explain transitive dependency in your own words.

3. Explain functional dependency in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.